Humidity and Temperature Controlled Testing Chamber

Junior Design – Group 3

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Testing Chamber Background

- Purpose:
  - Simulate environmental conditions to test electronic devices and materials in high temperature and humidity while remaining as accurate and user-friendly as possible.
- Description:
  - An insulated wooden box that utilizes an internal heater and humidifier system, which is controlled by a programmed Arduino Uno, and is powered with 120 V AC wall power ran through two power converters. The Arduino is controlled through a button that allows the user to select their chosen conditions.

Final Block Diagram

The TC block diagram shows how wall power flows through the power supply and buck converter to the Arduino, which reads the DHT22 sensor and automatically controls the heater, fans, and swamp cooler through three relays based on user setpoints adjusted via the rotary encoder and displayed on the LCD.

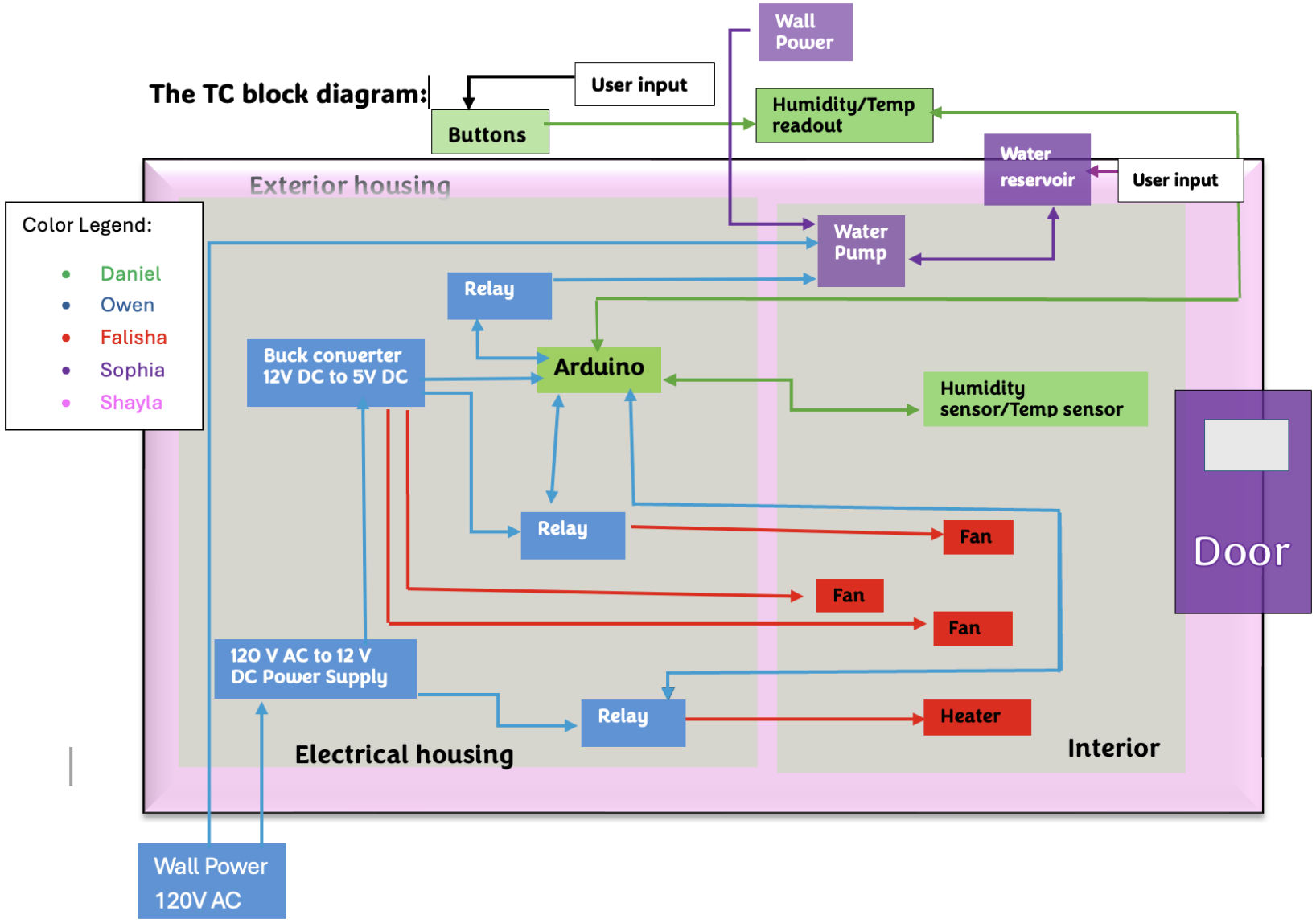


Fig. 1 Finalized TC Block Diagram

Finalized Housing CAD (via Onshape)

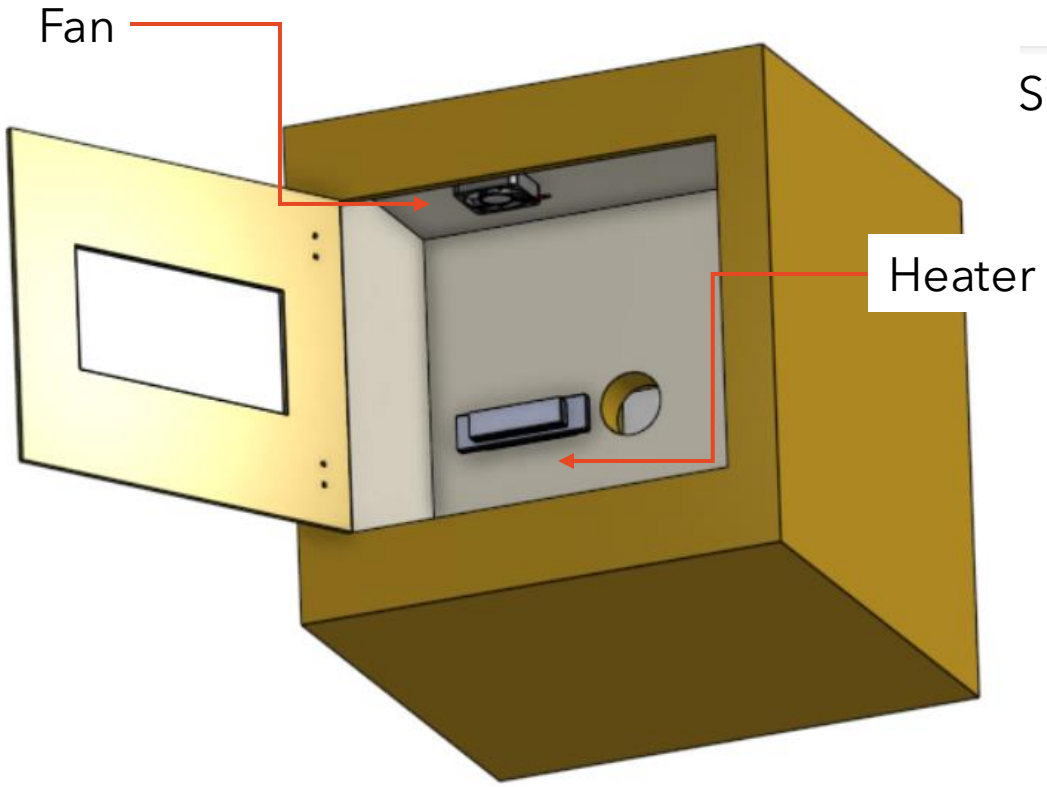


Fig. 2 Front CAD of TC housing

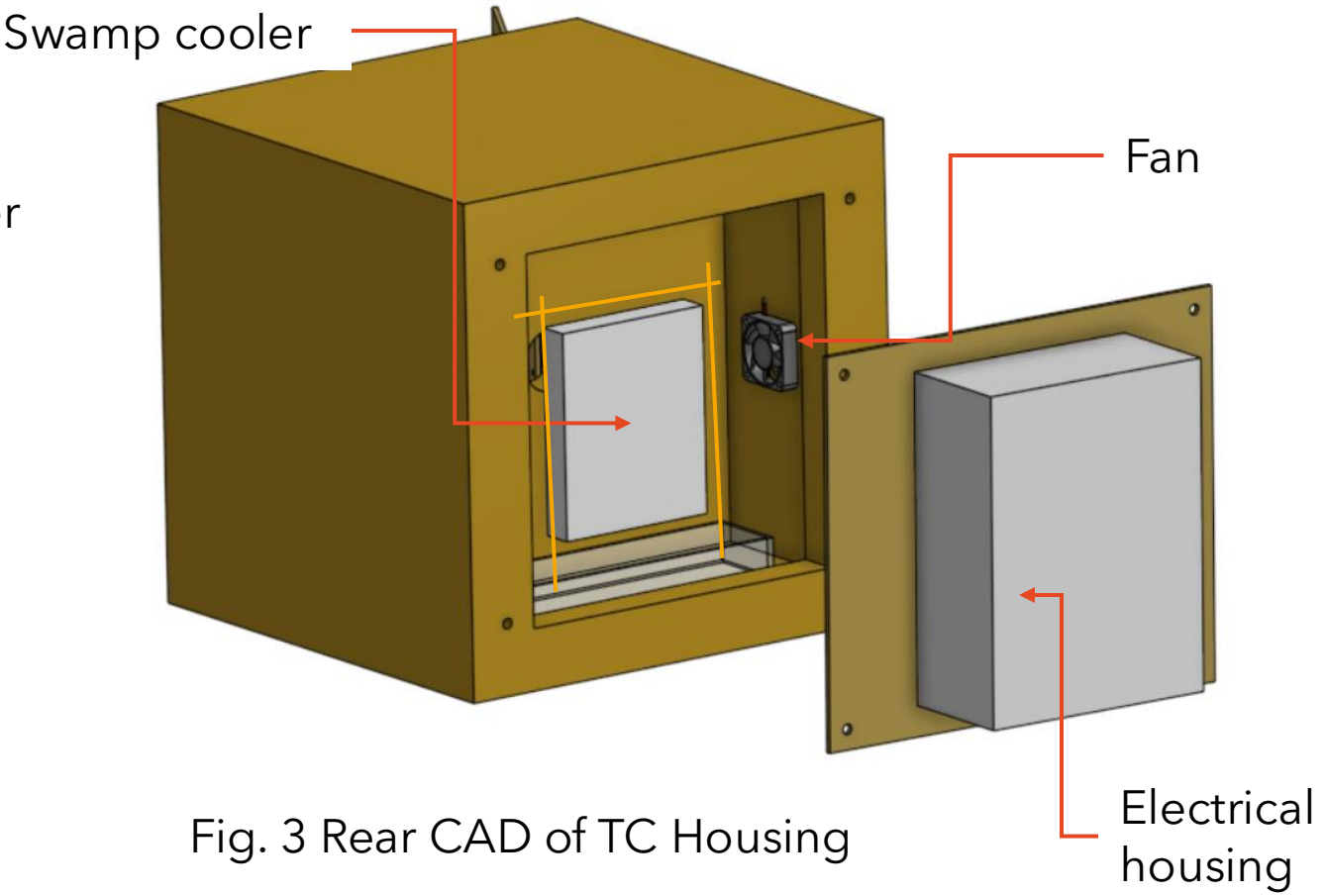
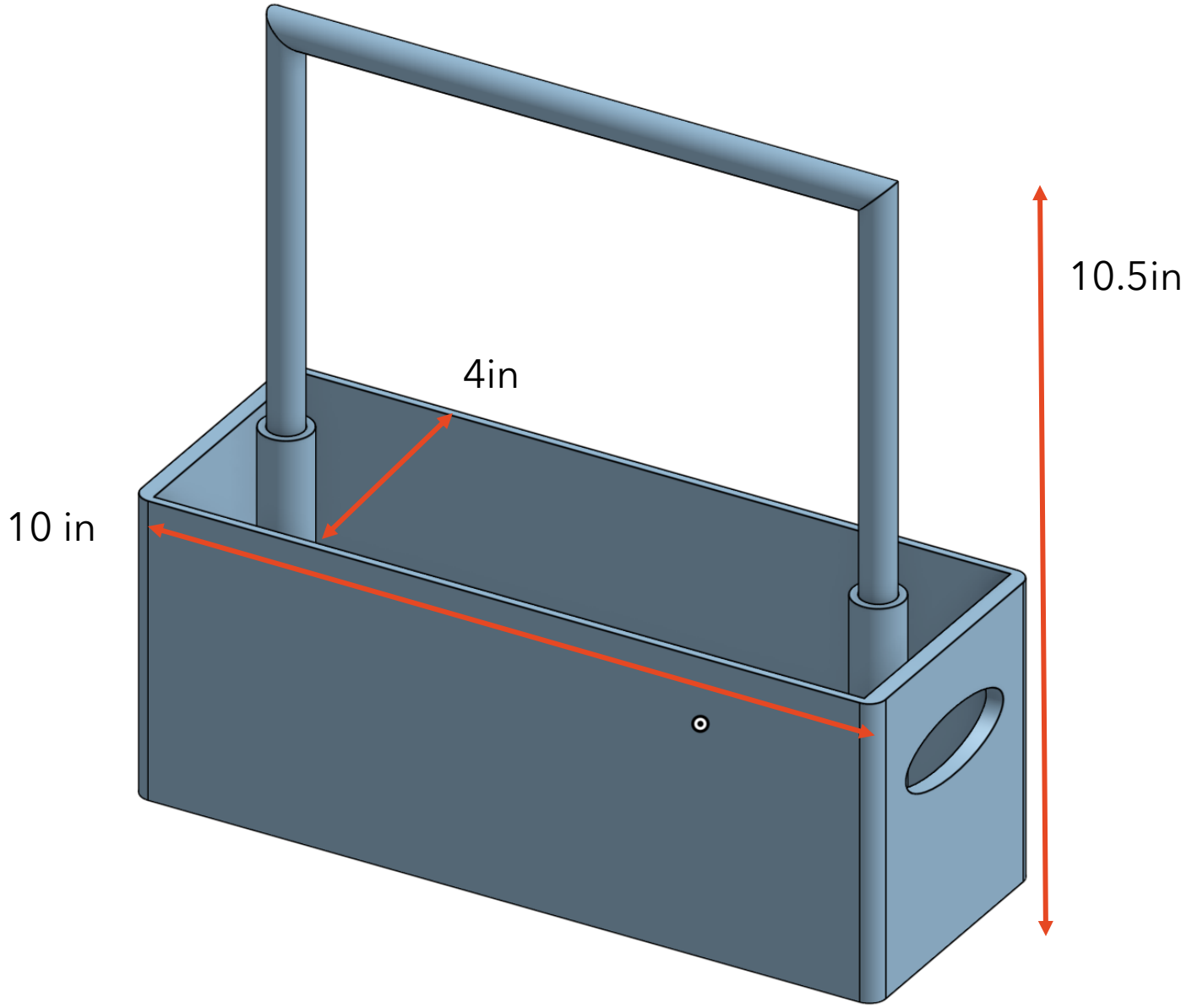
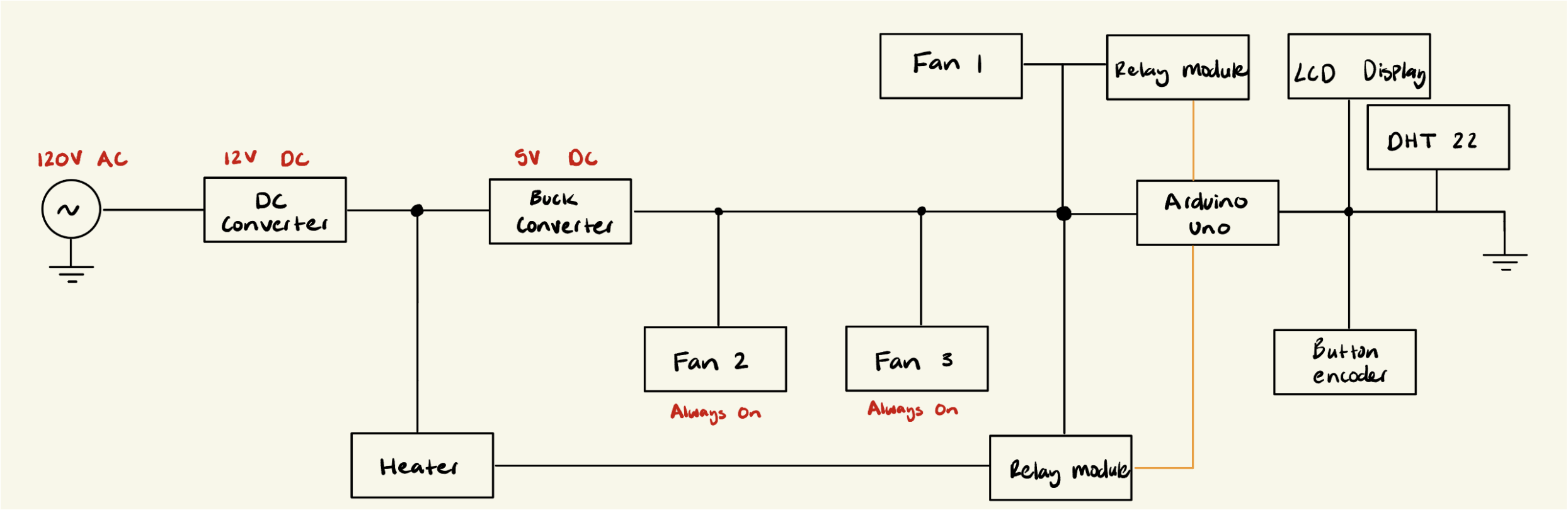


Fig. 3 Rear CAD of TC Housing

Finalized Swamp Cooler CAD



Electrical Schematic



Wokwi Simulation Electrical Schematic

- Reads temperature & humidity via DHT22 sensor
- Displays live data on a 16x2 LCD screen
- Automatically turns heater ON/OFF based on a set temperature
- 3 LEDs represent the heater, fan, and humidifier for simulation purposes.

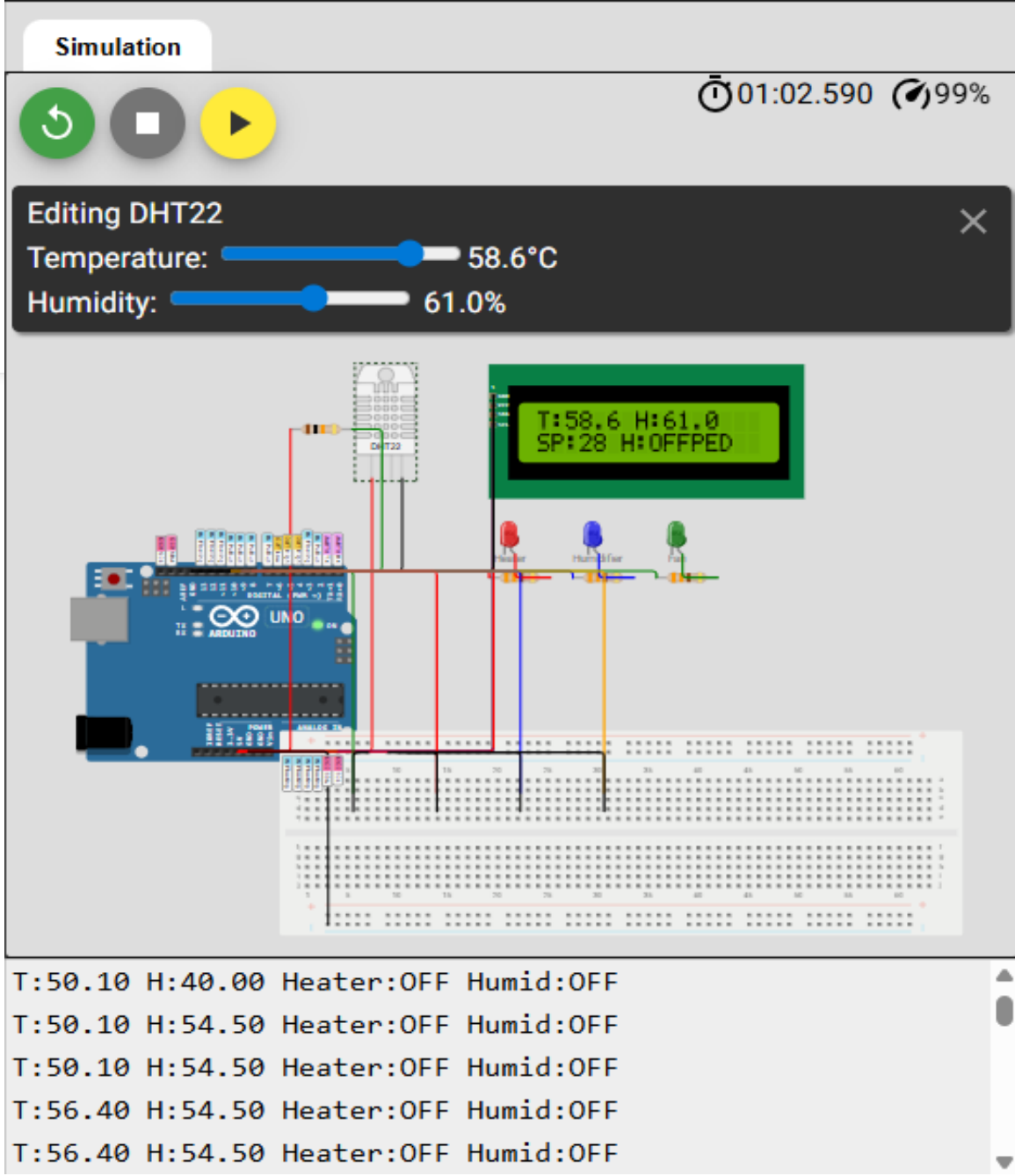


Fig. 4 Wokwi simulation screenshot

Power Budget Table

Component	Model	Voltage	Current (each)	Total Current	Power (W)
—— 5V CONTROL RAIL ——					
Arduino Uno R3	ATmega328P	5V	15–50 mA	1~50 mA	~0.25 W
DHT22 Sensor	AM2302	5V	1–1.5 mA	2~3 mA	~0.015 W
Relay Coil – Heater	SRD-05VDC-SL-C	5V	71.4 mA	171.4 mA	0.36 W
Relay Coil – Fan/Humidifier	SRD-05VDC-SL-C	5V	71.4 mA	171.4 mA	0.36 W
LCD Display	16x2 LCD	5V	~20 mA	120 mA	~0.10 W
Fan	CG4010H05S	5V	~0.18 A	3~0.36 A	~1.8 W
—— 12V RAIL ——					
Buck Converter loss	LM2596	12V→5V	–	1~200 mA	~0.50 W
Heater	PHMCR1D20/PH04-YT	120V AC	~4.17 A	1~4.17 A	~50 W
—— 120V AC LOADS ——					
Humidifier (Swamp Cooler)	–	120V AC	~0.21 A	1~0.21 A	~25 W

Rail	Total Current	Total Power
5V Rail Subtotal	~236 mA	~2.89 W
12V Rail Subtotal	~200 mA	~50.5 W
120V AC Loads Subtotal	~4.74 A	~25 W
TOTAL	~5.18 A	~78.39 W

Test Plan Results (Summary)

User Interface Specifications

- The system shall have a display showing the set temperature and humidity levels, and should have a display showing the set temperature and humidity levels along with a visual scale for easy viewing.
  - The LCD displays the set temperature and humidity levels when selecting, which is passing the *shall* spec.
- The materials will be visible in the chamber.
  - The door to the chamber has a plexiglass window, and the materials are visible through it.
- The system shall utilize user input to raise and lower the temperature and humidity set points, and it should utilize user input to raise and lower the temperature and humidity set points while showing the current measured temperature and humidity.
  - The system allows the user to adjust the selected temperature and humidity set points, and displays the actively measured temperature and humidity inside of the chamber, which is passing the *should* spec.

Test Plan Results (Summary)

Performance and Capability Specifications

- The system shall have the ability to maintain a temperature of ~32°C(90°F), and should have the ability to maintain a temperature of ~40°C(105°F).
  - The system could not reach a temperature of 40°C, but bypassed and maintained a maximum of 34.6°C, which is passing the *shall* spec.
- The chamber shall have the ability to maintain a humidity of 70% (RH) and should have the ability to maintain a humidity of 80% (RH).
  - The maximum humidity level the system could reach and maintain was 47.2%, which is not passing the spec.
- The system shall measure the humidity value within ± 5% relative humidity (RH) and should measure the humidity value within ± 3% relative humidity (RH).
  - On average, the system measures within ± 3.7% RH, which is passing the *shall* spec.

Test Plan Results (Summary)

Performance and Capability Specifications Continued

- The system shall measure the chamber temperature within ± 3 °Celsius and should measure the chamber temperature within ± 1 °Celsius.
  - On average, the system measures within ± 1.64°C, which is passing the *shall* spec.
- The system shall be able to reach the desired temperature from room temperature in 15 minutes. The system should be able to reach the desired temperature from room temperature in 10 minutes.
  - From a room temperature of 28°C, the system took under 10 minutes to reach desired temperatures up to 32°C. For 33°C, the system took over 10 minutes, but under 15 minutes, which is passing the *shall* spec.

Test Plan Results (Summary)

Size Specifications

- The system's internal testing chamber shall have dimensions no more than 12 in x 12 in x 12in and should have dimensions no more than 10.5 in x 10.5 in x 10.5 in.
  - The internal testing chamber is measured to be 9.5 in x 6 in x 10 in, which is passing the *should* spec.
- The external dimensions shall not exceed more than 24 in x 24 in x 24 in and should not exceed more than 20 in x 20 in x 20 in.
  - The external housing is measured to be 12 in x 12 in x 12 in, which is passing the *should* spec.
- The weight of the system shall be less than 65 pounds and should be less than 45 pounds.
  - The system weighs 13 pounds, which is passing the *should* spec.

Test Plan Results (Summary)

Safety Specifications

- The device exposed surfaces shall not exceed 35°C(95°F) and should not exceed 29.5°C(85°F).
- Results: Throughout the measurements, the device exposed surfaces were measured at 28.5 to 30.5 °C, which passed this safety test.
- The internal chamber shall be constructed of materials rated to withstand the maximum operating temperature without melting, warping, or degrading, and should maintain structural integrity with a thermal safety margin of at least 20% above the maximum setpoint of 40.5°C(105°F). Insulating materials shall be rated to at least 150°C and should be rated to at least 200°C to prevent heat damage to the chamber walls and surrounding components.
- Results: The thermal safety margin of the plywood used is about 93°C(200°F), while the thermal safety rating of the insulation used is 74°C(165°F). The interior of the testing chamber shows no signs of degradation after operating at the maximum setpoint, which passes the *shall* spec.



Test Plan Results (Summary)

Safety Specifications Continued

- In case of emergency, the system shall shut down within 10seconds after the user initiates the action and should shut down within 20 seconds after user initiates the action of turning off the switch.
  - The system effectively shut down within 15 seconds after switching off the power switch, which passes the test.



Test Plan Results (Summary)

Construction, Installation, and Sustainability (as appropriate) Specifications

- The system will run on DC voltage through a converter, which will have an input of AC wall power.
  - The system uses wall power, the output voltage at the DC converter was 12.24 V and the output at the Buck converter was 5.32 V. This passes our spec.
- The system will have an input voltage of 120V AC
  - The system was observed to use input wall power using the cord, which passes the spec.

Test Plan Results (Summary)

Construction, Installation, and Sustainability (as appropriate) Specifications Continued

- The system shall have a maximum physical size of 24 in × 24 in × 24 in and shall not exceed a total weight of 60 lbs. The system should have a maximum physical size of 16 in × 16 in × 16 in and should not exceed a total weight of 45 lbs.
  - The system has a physical size of 12 in x 14 in x 12 in and a total weight of 13 pounds, which is passing the *should* spec.
- The system shall be plugged in and ready for use, with temperature and humidity controls calibrated by the user, within 35 minutes. The system should be plugged in and ready for use, with temperature and humidity controls calibrated by the user, within 20 minutes.
  - Average time was 9 minutes after test trials, which passes our spec

Technical Challenges and Solutions

Electrical Housing & Component Placement

Challenge: Determining placement of the DC converter, LCD screen, and rotary encoder within the chamber housing without interfering with internal components or reducing usability.

Solution: Mounted the DC, LCD, and rotary encoder converter on the side panel of the electrical housing to keep it accessible and isolated from heat.

Relay connection to the heater

Challenge: Initially assumed the relay module required 12V to operate, leading to incorrect power routing for the heater control circuit.

Solution: After research, confirmed that the SRD-05VDC-SL-C relay module coil only requires 5V to activate the internal electromagnet that opens and closes the contacts – the 12V heater is connected solely to the relay's COM and NO terminals, which are electrically isolated from the control side. This allowed the relay to be driven directly from the Arduino's 5V rail without any separate high voltage control signal.

Housing manufacturing

Challenge: After assembly, some panels were too tall for the top panel to sit flush.

Solution: We manufactured spacers for the panels that were shorter.

Takeaways and Lessons Learned

- Consistent and clear communication is a necessity.
- Allow spare time in case of accidents and difficulty with subsystem integration, even if everything is planned properly.
- Plan the design as much as possible in advance so that manufacturing and integration are smooth.
- Make a budgeting sheet in Excel that includes all required parts of the system.
- How to work well with team members and adapt to others strength and weaknesses.

Adherence to Schedule

- We adhered to the schedule for the most part
- Where we went over:
  - Ordering of all parts took longer than expected (5/25/26)
  - Housing CAD deadline (design changed slightly) (5/30/26)
  - System integration/full assembly took a little longer because the wiring was too short (5/30/26)
- We had wiggle room in our schedule, so there was no real problems associated with the delays

Working Prototype Demo

Thanks For Watching!